

SAVVYDFIR-MCP Investigation Report

ROCBA-2020-FREDS-LAPTOP · Generated 2026-06-08T02:00:20.929016+00:00 ·

Summary

Investigation **ROCBA-2020-FREDS-LAPTOP** reached triage status **COMPLETE_WITH_GAPS** with **2** structurally confirmed finding(s), 2 of which are corroborated by multiple independent artifact sources, and **125** active lead(s) pending validation.

Confirmed activity (full detail in Top Confirmed Findings below):

- Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry, confidence 0.90) confirms the attacker's Outlook PST email exfiltration to Google Drive...
- Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch, confidence 0.95) confirms the attacker's active SRL IP data collection and physical...

Hunt closed: 2 confirmed (pre_planned_physical_intrusion_staged_account; multi_channel_exfiltration_usb_cloud_lateral_movement); 3 suspended (bitlocker_recovery_key_exfiltration; icloud_opportunistic_cloud_sync_exfiltration; anti_forensics_evidence_destruction_multi_method).

Cross-artifact synthesis: the confirmed activity is corroborated across multiple independent artifact sources.

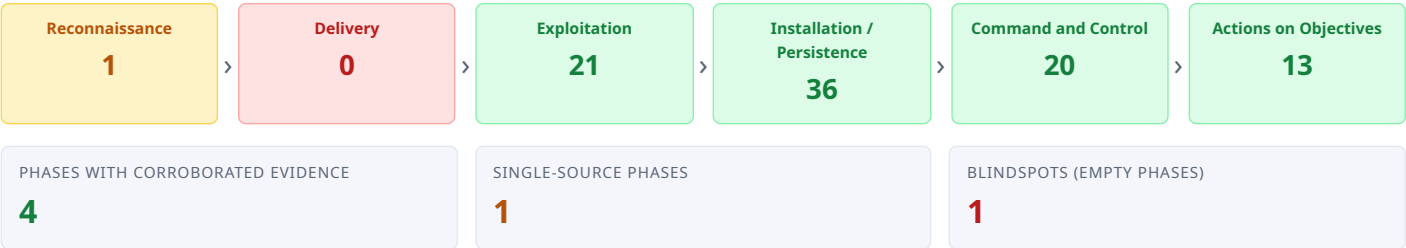
Kill-chain coverage: 5 of 6 phases evidenced, 1 blindspot(s).

Caveats: 5 unresolved cross-artifact discrepancy(ies).

CASE STATUS COMPLETE	TRIAGE COMPLETE_WITH_GAPS	FINDINGS 127	CONFIRMED 2	UNCONFIRMED FINDINGS 125	CORRECTION EVENTS 5
UNRESOLVED 5	SIGMA HITS 1	ATT&CK COVERAGE 35.7%			

Activity Thread - Cyber Kill Chain Coverage

Per Diamond Model Axiom 4, every malicious activity traverses a succession of kill-chain phases. Empty phases below represent **blindspots**: areas where evidence was either not extracted, not retained, or actively destroyed by anti-forensics. Phase color reflects the number of findings mapped to that phase.



KILL-CHAIN PHASE	STATUS
Reconnaissance	1 finding (single source)
Delivery	BLINDSPOT - no evidence
Exploitation	21 findings
Installation / Persistence	36 findings
Command and Control	20 findings
Actions on Objectives	13 findings

ATT&CK Coverage

Tactic-level coverage (covered / gap) is authoritative. Technique and finding columns are inferred from the findings mapped to each tactic.

TACTIC	STATUS	TECHNIQUES OBSERVED	FINDINGS	SUGGESTED NEXT TOOLS
TA0002 Execution	covered	-	-	
TA0003 Persistence	covered	-	-	
TA0005 Defense Evasion	covered	-	-	
TA0008 Lateral Movement	covered	-	-	
TA0011 Command and Control	covered	-	-	
TA0001 Initial Access	gap	-	-	disk.summarize_evtx, memory.scan_network
TA0004 Privilege Escalation	gap	-	-	disk.summarize_evtx, memory.scan_processes
TA0006 Credential Access	gap	-	-	disk.extract_shimcache, disk.get_amcache, disk.summarize_evtx
TA0007 Discovery	gap	-	-	memory.scan_processes, memory.list_dlls, timeline.query_timeline
TA0009 Collection	gap	-	-	disk.extract_prefetch, disk.get_amcache, memory.scan_processes
TA0010 Exfiltration	gap	-	-	disk.extract_srum, memory.scan_network, timeline.query_timeline
TA0040 Impact	gap	-	-	disk.summarize_evtx, memory.detect_injection, timeline.query_timeline
TA0042 Resource Development	gap	-	-	disk.get_amcache, disk.extract_prefetch, timeline.query_timeline
TA0043 Reconnaissance	gap	-	-	disk.summarize_evtx, memory.scan_network, timeline.query_timeline

Key Event Timeline

TIME (UTC)	FINDING	STATUS	EVENT
2020-10-21T03:44:22Z	F-059	ACTIVE	Amcache confirms the srl-h account downloaded SDelete (sdelete64.exe, SHA-1:...
2020-10-21T03:46:19Z	F-118	ACTIVE	srl-h's PowerShell history records sequential execution of sdelete64.exe -nobanner -z -c D: and sdelete64.exe...
2020-10-21T03:46:19Z	F-117	ACTIVE	srl-h PowerShell history (mtime 2020-10-21T03:46:19 UTC) records 6 commands: initial attempts to run sdelete64.exe...
2020-11-13T00:00:01Z	F-047	ACTIVE	NTLM brute force campaign: 58,713 EID 4776 (NTLM validation) and 58,706 EID 4625 (failed logon) events in...
2020-11-14T00:34:00Z	F-122	ACTIVE	Chainsaw Sigma scan (2,284 rules, E-201) against Security.evtx, System.evtx, PowerShell.evtx, and Application.evtx...
2020-11-14T03:42:49Z	F-124	CONFIRMED	Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch,...
2020-11-14T03:42:49Z	F-048	ACTIVE	INBOUND RDP takeover from attacker host 'cobra' (52.249.198.56, Azure IP): At 2020-11-14 03:42:44 UTC (22:42 EST Nov...
2020-11-14T03:43:00Z	F-102	ACTIVE	LNK file analysis confirms the breadth of SRL intellectual property accessed and exfiltrated during the 2020-11-14...
2020-11-14T03:46:00Z	F-101	ACTIVE	CRIMSON2 (F:) USB drive was used as primary physical exfiltration medium. LNK files created during attack window...
2020-11-14T03:48:53Z	F-106	ACTIVE	Jump Lists confirm the attacker accessed Maria Hill's (mhill) private SharePoint personal documents through Fred's...
2020-11-14T04:13:22Z	F-109	ACTIVE	Edge browser download record at 04:13:22 UTC proves a file from Maria Hill's private SharePoint OneDrive was...
2020-11-14T05:00:44Z	F-049	ACTIVE	Lateral movement from Fred's laptop to shieldbase infrastructure: At 2020-11-14 05:00:44 UTC (00:00 EST) attacker...
2020-11-14T05:05:38Z	F-104	ACTIVE	Jump List for Remote Desktop Connection (AppId: 1bc392b8e104a00e) proves mstsc.exe was used to RDP to...
2020-11-14T12:48:13Z	F-114	ACTIVE	Recycle Bin entry \$IDNBREY.pst proves the BitLocker recovery key file was found on Fred's OneDrive Desktop and...
2020-11-14T13:37:51Z	F-108	ACTIVE	Browser history (Chrome) proves SDelete was searched and downloaded by the attacker during the 2020-11-14 attack...
2020-11-14T13:39:21Z	F-052	ACTIVE	Attacker created backup.pst at 2020-11-14 13:39:21 UTC in Users\fredr\OneDrive\Documents\Outlook Files, wrote email...
2020-11-14T13:42:30Z	F-056	ACTIVE	SDELETE.EXE executed 7 times across two paths during the attacker's session (13:42-13:47 UTC 2020-11-14). First 2...
2020-11-14T14:00:50Z	F-125	CONFIRMED	Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry,...
2020-11-14T14:00:50Z	F-096	ACTIVE	ShellBags confirm Google Drive File Stream was mounted as G:\ and the attacker actively browsed the following G:...
2020-11-14T14:01:34Z	F-105	ACTIVE	Jump Lists confirm Microsoft Outlook 2016 was used to access both the local PST file (backup.pst) and the Google...
2020-11-14T14:01:35Z	F-100	ACTIVE	LNK file SRL-EMAIL-EXPORT.lnk (SourceCreated 2020-11-14 14:01:35 UTC) definitively proves the Outlook PST export was...
2020-11-14T14:04:07Z	F-112	ACTIVE	WordWheelQuery registry key (NTUSER.DAT, LastWriteTimestamp 2020-11-14 14:04:07 UTC) records 11 search terms typed...

TIME (UTC)	FINDING	STATUS	EVENT
2020-11-14T14:07:32Z	F-115	ACTIVE	Recycle Bin contains four post-exfiltration cleanup events establishing the attacker's cleanup sequence: (1)...

Top Confirmed Findings

F-125 CONFIRMED conf 0.97 (high) / T1114.001

Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry, confidence 0.90) confirms the attacker's Outlook PST email exfiltration to Google Drive cloud. Three independent artifact streams co-occur within 4 minutes: EVTX records...

Corroborated by F-050, F-105, F-112, F-100

execution_id E-206 · **evidence** inference · **disposition** ruled_out · **alt. hypothesis** Fred performed a routine Outlook PST backup operation before departing on vacation, and the registry searches reflect his normal maintenance workflow. · **ruled out by** Fred departed on 2020-11-10 (manifest), five days before this 2020-11-14 activity; the search sequence *.pst → backup → backup.pst (F-112) proves the PST...

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst

file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst

timestamp: 2020-11-14T14:01:34

timestamp: 2020-11-14T14:03:52

timestamp: 2020-11-14T14:04:07

F-124 CONFIRMED conf 0.95 (high) / T1005

Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch, confidence 0.95) confirms the attacker's active SRL IP data collection and physical exfiltration window. Four independent artifact streams co-occur: EVTX records...

Corroborated by F-048, F-043, F-101, F-102

execution_id E-206 · **evidence** inference · **disposition** ruled_out · **alt. hypothesis** Fred or another authorized SRL user remotely accessed the laptop during this window and performed routine file operations. · **ruled out by** Fred was confirmed absent on vacation since 2020-11-10 (manifest); srl-h account (not Fred's account fredr) was the active session per EVTX logon records;...

timestamp: 2020-11-14T03:42:49

timestamp: 2020-11-14T03:46:33

executable: CRIMSON2 USB volume mounted

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\SRL-Projects

Active Leads & Recommended Pivots

These are **unconfirmed** findings - single-source observations and leads to pivot on, not corroborated conclusions. Only items in "Top Confirmed Findings" above meet the multi-source CONFIRMED bar.

SEVERITY	DETECTOR	CONFIDENCE	DESCRIPTION	RECOMMENDED NEXT PIVOT
MEDIUM	motw_presence	0.70	Mark-of-the-Web Zone.Identifier evidence is present.	Correlate MotW file creation with browser, LNK, Jump List, and execution artifacts.

Active Finding Leads

F-115 ACTIVE conf 0.99 (high) / T1070.004

Recycle Bin contains four post-exfiltration cleanup events establishing the attacker's cleanup sequence: (1) backup.pst (\$IDNBREY.pst) deleted at 2020-11-14 14:07:32 UTC — original path C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst, size 20,587,520 bytes (20.6...

file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst

file_path: C:\Users\fredr\Downloads\SDelete\sdelete.exe

timestamp: 2020-11-14T14:07:32Z

timestamp: 2020-11-14T13:41:24Z

timestamp: 2020-11-14T04:49:53Z

F-114 ACTIVE conf 0.99 (high) / T1552.001

Recycle Bin entry \$IDNBREY.pst proves the BitLocker recovery key file was found on Fred's OneDrive Desktop and deleted to the Recycle Bin at 2020-11-14 12:48:13 UTC. Original path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key...

file_path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key 26F77152-999C-45E8-8BD4-C83FAC7BB72D.TXT

timestamp: 2020-11-14T12:48:13Z

F-109 ACTIVE conf 0.99 (high) / T1213.002

Edge browser download record at 04:13:22 UTC proves a file from Maria Hill's private SharePoint OneDrive was downloaded directly to the CRIMSON2 USB drive. Source URL:...

file_path: F:\Files from SRL system\Quantum Particles Affected by Other Dimensions.pdf

file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/Quantum%20Particles%20Affected%20by%20Other%20Dimensions.pdf

timestamp: 2020-11-14T04:13:22Z

F-108 ACTIVE conf 0.99 (high) / T1070.004

Browser history (Chrome) proves SDelete was searched and downloaded by the attacker during the 2020-11-14 attack session. At 13:37:37 UTC Chrome shows Google search for "sdelete download"; at 13:37:47 visited docs.microsoft.com/sysinternals/downloads/sdelete; two download...

executable: C:\Users\fredr\Downloads\SDelete.zip

file_path: C:\Users\fredr\Downloads\SDelete.zip

timestamp: 2020-11-14T13:37:51Z

timestamp: 2020-11-14T13:37:37Z

F-105 ACTIVE conf 0.99 (high) / T1114.001

Jump Lists confirm Microsoft Outlook 2016 was used to access both the local PST file (backup.pst) and the Google Drive PST (SRL-EMAIL-EXPORT.pst). Outlook AppId (Microsoft Outlook 2016 64-bit) entry for backup.pst: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst,...

file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst

timestamp: 2020-11-14T14:01:34Z

timestamp: 2020-11-14T14:03:52Z

timestamp: 2020-11-14T14:00:54Z

F-056 ACTIVE conf 0.98 (high) / T1070.004

SDELETE.EXE executed 7 times across two paths during the attacker's session (13:42-13:47 UTC 2020-11-14). First 2 runs from Users\Fredr\Downloads\SDelete\SDELETE.EXE (Prefetch hash -2BD91720) at 13:42:30 and 13:42:38 UTC; subsequent 5 runs with SDelete copied to...

executable: SDELETE.EXE file_path: Users\Fredr\Downloads\SDelete\SDELETE.EXE file_path: Windows\System32\SDELETE.EXE
file_path: Users\Fredr\OneDrive\Desktop\EARTHFORCE SA-26 THUNDERBOLT STAR FURY.DOCX file_path: Users\Fredr\OneDrive\ADAMANTIUM-BACKGROUND.DOCX
file_path: Users\Fredr\OneDrive\SUCCESS-TEST-PLAN-VIBRANIUM-ALLOY-RESULTS.DOCX

F-112 ACTIVE conf 0.98 (high) / T1083

WordWheelQuery registry key (NTUSER.DAT, LastWriteTimestamp 2020-11-14 14:04:07 UTC) records 11 search terms typed into Windows Explorer's search bar, revealing the attacker's systematic intelligence-collection methodology in reverse MRU order: MRU:10 starfury, MRU:9 kitt,...

timestamp: 2020-11-14T14:04:07Z si_modified: 2020-11-14T14:04:07Z

F-100 ACTIVE conf 0.98 (high) / T1567.002

LNK file SRL-EMAIL-EXPORT.lnk (SourceCreated 2020-11-14 14:01:35 UTC) definitively proves the Outlook PST export was placed on Google Drive File Stream as G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst (VolumeLabel: Google Drive File Stream,...

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst
file_path: C:\Users\Fredr\OneDrive\Documents\Outlook Files\backup.pst timestamp: 2020-11-14T14:01:35Z timestamp: 2020-11-14T14:03:52Z

F-048 ACTIVE conf 0.98 (high) / T1021.001

INBOUND RDP takeover from attacker host 'cobra' (52.249.198.56, Azure IP): At 2020-11-14 03:42:44 UTC (22:42 EST Nov 13) the LOCAL console session for SRL-FORGE\Fredr was disconnected (EID 4779, Address: LOCAL) and within 5 seconds was reconnected via RDP from hostname...

owner_process: 52.249.198.56 process_path: cobra timestamp: 2020-11-14T03:42:49

F-123 ACTIVE conf 0.97 (high) / T1070.004

compare_disk_and_memory (E-204) confirms two execution_evidence_deleted_binary HIGH discrepancies for SDELETE.EXE (F-056/F-115) and sdelete64.exe (F-059/F-118): Prefetch/Amcache proves both binaries executed; Recycle Bin and PowerShell history prove both were subsequently...

executable: SDELETE.EXE executable: sdelete64.exe file_path: C:\Users\Fredr\Downloads\SDelete\sdelete.exe timestamp: 2020-11-14T13:41:24Z

F-106 ACTIVE conf 0.97 (high) / T1213.002

Jump Lists confirm the attacker accessed Maria Hill's (mhill) private SharePoint personal documents through Fred's authenticated Office 365 session, opening 5 files from mhill's SharePoint site: (1)...

file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/KITT-CompetitiveAnalysisDocs/The%20Future%20of%20KITT.pptx
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/WorkingFiles/German-KITT-Specs.docx
timestamp: 2020-11-14T03:48:53Z timestamp: 2020-11-14T04:23:05Z

F-104 ACTIVE conf 0.97 (high) / T1021.001

Jump List for Remote Desktop Connection (AppId: 1bc392b8e104a00e) proves mstsc.exe was used to RDP to "base-rd-08.shieldbase.lan" on 2020-11-14, LastModified 05:05:38 UTC with InteractionCount=4 (connected 4 times). The DestList Arguments field contains...

executable: C:\Windows\System32\mstsc.exe owner_process: mstsc.exe timestamp: 2020-11-14T05:05:38Z path: base-rd-08.shieldbase.lan

F-101 **ACTIVE** **conf 0.97 (high)** / T1052.001

CRIMSON2 (F:\) USB drive was used as primary physical exfiltration medium. LNK files created during attack window 2020-11-14 03:46–04:30 UTC confirm the attacker brought the CRIMSON2 drive (VolumeLabel: CRIMSON2) and structured it with pre-planned folders: F:\Files from SRL...

file_path: F:\Files from SRL system file_path: F:\Files of interest\Megaforce file_path: F:\Key Data\Blue Thunder
file_path: F:\Recovered Documents\Wolves_Lair_Tech_Specs.pptx timestamp: 2020-11-14T03:46:00Z timestamp: 2020-11-14T04:30:00Z

F-049 **ACTIVE** **conf 0.97 (high)** / T1021.001

Lateral movement from Fred's laptop to shieldbase infrastructure: At 2020-11-14 05:00:44 UTC (00:00 EST) attacker used RDP Client to connect to base-rd-08.shieldbase.lan (172.16.6.18, private IP). TerminalServices-RDPClient EID 1024 (Dest: base-rd-08.shieldbase.lan) and EID...

owner_process: base-rd-08.shieldbase.lan process_path: 172.16.6.18 value_data: SHIELDBASE\frocba timestamp: 2020-11-14T05:00:44

F-102 **ACTIVE** **conf 0.96 (high)** / T1005

LNK file analysis confirms the breadth of SRL intellectual property accessed and exfiltrated during the 2020-11-14 attack. Under fredr's profile, attacker accessed (ordered by time UTC): ADAMANTIUM-Background.docx (03:43, C:\Users\fredr\Documents\SRL);...

file_path: C:\Users\fredr\Documents\SRL\ADAMANTIUM-Background.docx
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Research to Weaponize the Ion Thruster.docx
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air\Wolf Air Financials.xlsx
file_path: C:\Users\fredr\OneDrive\Documents\Vibrainium-SRL.docx timestamp: 2020-11-14T03:43:00Z timestamp: 2020-11-14T04:36:51Z

F-127 **ACTIVE** **conf 0.95 (high)** / T1490

analyze_vss (E-210) found 0 Volume Shadow Copies on the rocba-cdrive.e01 image. VSS is absent, meaning pre-clearing Security.evtx recovery is not possible. The absence may reflect: (1) VSS was disabled on this Windows 10 device (common on devices with small SSD storage), (2)...

timestamp: 2020-11-14T00:34:00 executable: vshadowinfo confirmed 0 shadow stores

F-059 **ACTIVE** **conf 0.95 (high)** / T1136.001

Amcache confirms the srl-h account downloaded SDelete (sdelete64.exe, SHA-1: 7bcd946326b67f806b3db4595ede9fbdf29d0c36) to c:\users\srl-h\Downloads\ on 2020-10-21 03:44:22 UTC — 24 days before the break-in. The identical SHA-1 appears for...

executable: sdelete64.exe file_path: c:\users\srl-h\downloads\sdelete64.exe file_path: c:\users\fredr\downloads\sdelete\sdelete.exe
timestamp: 2020-10-21T03:44:22Z timestamp: 2020-11-14T13:42:33Z

F-122 **ACTIVE** **conf 0.95 (high)** / T1070.001

Chainsaw Sigma scan (2,284 rules, E-201) against Security.evtx, System.evtx, PowerShell.evtx, and Application.evtx produced 0 medium/high/critical detections. All 7 hits are "User Logoff Event" (T1531, informational) timestamped 2020-11-16 02:29:36-42 UTC (forensic team...

timestamp: 2020-11-14T00:34:00 executable: Security.evtx chunk parse failures chunks 254-289

F-096 **ACTIVE** **conf 0.95 (high)** / T1567.002

ShellBags confirm Google Drive File Stream was mounted as G:\ and the attacker actively browsed the following G: drive folders during the attack: G:\My Drive\STARK-RESEARCH-LABS FOLDER\ (03:57:03 UTC), G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air (04:27:21...

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER timestamp: 2020-11-14T14:00:50Z timestamp: 2020-11-14T03:57:03Z

F-052 ACTIVE conf 0.95 (high) / T1114.001

Attacker created backup.pst at 2020-11-14 13:39:21 UTC in Users\fredr\OneDrive\Documents\Outlook Files, wrote email data to it until 14:02 UTC, then deleted it to the Recycle Bin (\$RDNBREY.pst) at 14:07:33 UTC. USN confirms FileCreate → DataOverwrite sequence followed by...

`file_path: Users\fredr\OneDrive\Documents\Outlook Files\backup.pst``file_path: $Recycle.Bin\RDNBREY.pst``file_path: $Recycle.Bin\IDNBREY.pst``timestamp: 2020-11-14T13:39:21Z``timestamp: 2020-11-14T14:07:33Z`**F-047** ACTIVE conf 0.95 (high) / T1110.003

NTLM brute force campaign: 58,713 EID 4776 (NTLM validation) and 58,706 EID 4625 (failed logon) events in Security.evtx spanning 2020-11-13 to 2020-11-14 UTC. Targets include ADMINISTRATOR (0xC000006A = correct username, wrong password — account confirmed to exist), and...

`executable: ADMINISTRATOR``timestamp: 2020-11-13T00:00:01`**F-118** ACTIVE conf 0.92 (high) / T1070.004

srl-h's PowerShell history records sequential execution of sdelete64.exe -nobanner -z -c D: and sdelete64.exe -nobanner -z -c C:, zeroing all free space on both C: and D: drives to frustrate file-recovery forensics. No fredr PowerShell history exists, confirming the break-in...

`executable: sdelete64.exe``process_path: C:\Users\srl-h\Downloads\sdelete64.exe``timestamp: 2020-10-21T03:46:19`**F-117** ACTIVE conf 0.92 (high) / T1021.001

srl-h PowerShell history (mtime 2020-10-21T03:46:19 UTC) records 6 commands: initial attempts to run sdelete64.exe via UNC tsclient path \\tsclient\C\Users\rsydow\Documents\SRL Admin\Workstation-Prep\sdelete64.exe targeting D: drive, followed by cd to srl-h Downloads and...

`path: \\tsclient\C\Users\rsydow\Documents\SRL Admin\Workstation-Prep\sdelete64.exe``executable: sdelete64.exe``process_path: C:\Users\srl-h\Downloads\sdelete64.exe``timestamp: 2020-10-21T03:46:19`

Corroboration & Suggested Next Steps (advisory)

Advisory only - does not change finding_status or confidence. Distinct from ATT&CK *Suggested Next Tools* (tactic blind-spots) and *Active Leads* (detector pivots): this is forensic-knowledge corroboration - what would raise each finding to the next evidentiary tier.

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
F-001	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-002	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-003	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-004	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-005	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-006	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-007	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-008	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-009	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-010	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-011	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-012	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-013	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-014	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-015	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-016	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
						netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-017	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-018	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-019	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-020	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-021	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-022	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-023	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-024	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
F-025	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-026	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-027	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-028	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-029	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-030	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-031	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-032	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-033	volatility_memory	observation	unknown	memory netscan (PID->C2 socket),	memory.scan_processes	advisory only; does not reflect finding_status or promotion.

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				memory cmdline (encoded/hidden args)		memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-034	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-035	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-036	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-037	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-038	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-039	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-040	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-041	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
						source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-042	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-043	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-044	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-045	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-046	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-047	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-048	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-049	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-050	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.

Indicators of Compromise

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst	path/file	F-125	CONFIRMED	T1114.001
file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst	path/file	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:01:34	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:03:52	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:04:07	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T03:42:49	indicator	F-124	CONFIRMED	T1005
timestamp: 2020-11-14T03:46:33	indicator	F-124	CONFIRMED	T1005
executable: CRIMSON2 USB volume mounted	executable	F-124	CONFIRMED	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\SRL-Projects	path/file	F-124	CONFIRMED	T1005
file_path: C:\Users\fredr\Downloads\SDelete\sdelete.exe	path/file	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T14:07:32Z	indicator	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T13:41:24Z	indicator	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T04:49:53Z	indicator	F-115	ACTIVE	T1070.004
file_path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key 26F77152-999C-45E8-8BD4-C83FAC7BB72D.TXT	path/file	F-114	ACTIVE	T1552.001
timestamp: 2020-11-14T12:48:13Z	indicator	F-114	ACTIVE	T1552.001
file_path: F:\Files from SRL system\Quantum Particles Affected by Other Dimensions.pdf	path/file	F-109	ACTIVE	T1213.002
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/Quantum%20Particles%20Affected%20by%20Other%20Dimensions.pdf	indicator	F-109	ACTIVE	T1213.002
timestamp: 2020-11-14T04:13:22Z	indicator	F-109	ACTIVE	T1213.002
executable: C:\Users\fredr\Downloads\SDelete.zip	executable	F-108	ACTIVE	T1070.004
file_path: C:\Users\fredr\Downloads\SDelete.zip	path/file	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T13:37:51Z	indicator	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T13:37:37Z	indicator	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T14:01:34Z	indicator	F-105	ACTIVE	T1114.001
timestamp: 2020-11-14T14:03:52Z	indicator	F-105	ACTIVE	T1114.001
timestamp: 2020-11-14T14:00:54Z	indicator	F-105	ACTIVE	T1114.001
executable: SDELETE.EXE	executable	F-056	ACTIVE	T1070.004
file_path: Users\fredr\Downloads\SDelete\SDELETE.EXE	path/file	F-056	ACTIVE	T1070.004
file_path: Windows\System32\SDELETE.EXE	path/file	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\Desktop\EARTHFORCE SA-26 THUNDERBOLT STAR FURY.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\ADAMANTIUM-BACKGROUND.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\SUCCESS-TEST-PLAN-VIBRANIUM-ALLOY-RESULTS.DOCX	account	F-056	ACTIVE	T1070.004

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
file_path: Users\fredr\OneDrive\THE SHIELD BACKGROUND AND ONGOING RESEARCH.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\NOKIA STRATEGY.DOCX	account	F-056	ACTIVE	T1070.004
timestamp: 2020-11-14T13:42:30Z	indicator	F-056	ACTIVE	T1070.004
timestamp: 2020-11-14T14:04:07Z	indicator	F-112	ACTIVE	T1083
si_modified: 2020-11-14T14:04:07Z	indicator	F-112	ACTIVE	T1083
timestamp: 2020-11-14T14:01:35Z	indicator	F-100	ACTIVE	T1567.002
owner_process: 52.249.198.56	process	F-048	ACTIVE	T1021.001
process_path: cobra	indicator	F-048	ACTIVE	T1021.001
executable: sdelete64.exe	executable	F-123	ACTIVE	T1070.004
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/KITT-CompetitiveAnalysisDocs/The%20Future%20of%20KITT.pptx	indicator	F-106	ACTIVE	T1213.002
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/WorkingFiles/German-KITT-Specs.docx	indicator	F-106	ACTIVE	T1213.002
timestamp: 2020-11-14T03:48:53Z	indicator	F-106	ACTIVE	T1213.002
timestamp: 2020-11-14T04:23:05Z	indicator	F-106	ACTIVE	T1213.002
executable: C:\Windows\System32\mstsc.exe	executable	F-104	ACTIVE	T1021.001
owner_process: mstsc.exe	process	F-104	ACTIVE	T1021.001
timestamp: 2020-11-14T05:05:38Z	indicator	F-104	ACTIVE	T1021.001
path: base-rd-08.shieldbase.lan	indicator	F-104	ACTIVE	T1021.001
file_path: F:\Files from SRL system	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Files of interest\Megaforce	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Key Data\Blue Thunder	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Recovered Documents\Wolves_Lair_Tech_Specs.pptx	path/file	F-101	ACTIVE	T1052.001
timestamp: 2020-11-14T03:46:00Z	indicator	F-101	ACTIVE	T1052.001
timestamp: 2020-11-14T04:30:00Z	indicator	F-101	ACTIVE	T1052.001
owner_process: base-rd-08.shieldbase.lan	process	F-049	ACTIVE	T1021.001
process_path: 172.16.6.18	ip	F-049	ACTIVE	T1021.001
value_data: SHIELDBASE\frocba	account	F-049	ACTIVE	T1021.001
timestamp: 2020-11-14T05:00:44	indicator	F-049	ACTIVE	T1021.001
file_path: C:\Users\fredr\Documents\SRL\ADAMANTIUM-Background.docx	path/file	F-102	ACTIVE	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Research to Weaponize the Ion Thruster.docx	path/file	F-102	ACTIVE	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air\Wolf Air Financials.xlsx	path/file	F-102	ACTIVE	T1005
file_path: C:\Users\fredr\OneDrive\Documents\Vibrainium-SRL.docx	path/file	F-102	ACTIVE	T1005
timestamp: 2020-11-14T03:43:00Z	indicator	F-102	ACTIVE	T1005

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
timestamp: 2020-11-14T04:36:51Z	indicator	F-102	ACTIVE	T1005
timestamp: 2020-11-14T00:34:00	indicator	F-127	ACTIVE	T1490
executable: vshadowinfo confirmed 0 shadow stores	executable	F-127	ACTIVE	T1490
file_path: c:\users\srl-h\downloads\sdelete64.exe	path/file	F-059	ACTIVE	T1136.001
file_path: c:\users\fredr\downloads\sdelete\sdelete.exe	path/file	F-059	ACTIVE	T1136.001
timestamp: 2020-10-21T03:44:22Z	indicator	F-059	ACTIVE	T1136.001
timestamp: 2020-11-14T13:42:33Z	indicator	F-059	ACTIVE	T1136.001
executable: Security.evtx chunk parse failures chunks 254-289	executable	F-122	ACTIVE	T1070.001
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST	path/file	F-096	ACTIVE	T1567.002
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air	path/file	F-096	ACTIVE	T1567.002
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER	path/file	F-096	ACTIVE	T1567.002
timestamp: 2020-11-14T14:00:50Z	indicator	F-096	ACTIVE	T1567.002
timestamp: 2020-11-14T03:57:03Z	indicator	F-096	ACTIVE	T1567.002
file_path: Users\fredr\OneDrive\Documents\Outlook Files\backup.pst	account	F-052	ACTIVE	T1114.001
file_path: \$Recycle.Bin\SRDNBREY.pst	account	F-052	ACTIVE	T1114.001
file_path: \$Recycle.Bin\IDNBREY.pst	account	F-052	ACTIVE	T1114.001
timestamp: 2020-11-14T13:39:21Z	indicator	F-052	ACTIVE	T1114.001

Anti-Forensics Warnings

- None recorded.

Data Gaps

- **warning:** Terminal absence: 0 shadow copies on rocba-cdrive.e01 — pre-clearing Security.evtx events (before 2020-11-14 00:34 UTC) are permanently irrecoverable; attacker log clearing, initial access, and credential-theft events cannot be recovered
- **warning:** Cannot determine whether VSS absence is from configuration, storage limits, or attacker deletion — EID 4688 for vssadmin/wmic unavailable due to cleared Security.evtx

TECHNICAL APPENDIX

Sigma Anomaly Summary

#	Severity	Detector	Description	ATT&CK
1	MEDIUM	motw_presence	Mark-of-the-Web Zone.Identifier evidence is present.	T1204

Full Finding Index (127)

Every recorded finding, so any F-NNN can be looked up from this report alone. CONFIRMED = multi-source corroborated conclusion; ACTIVE/OBSERVATION = an unconfirmed lead (a single-source observation, not yet a conclusion).

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-001	ACTIVE	0.70	hypothesis	T1036	memory.list_processes	E-005	Suspicious process: csrss.exe (PID 800). Reason: cs got 'svchost.exe'. Path: unknown. PPID: 740 (svcho
F-002	ACTIVE	0.70	hypothesis	T1036	memory.list_processes	E-005	Suspicious process: winlogon.exe (PID 896). Reaso 'smss.exe' but got 'svchost.exe'. Path: unknown. PF
F-003	ACTIVE	1.00	observation		memory.list_processes	E-005	Pslist baseline: 2186 active processes from /eviden
F-004	ACTIVE	0.90	observation		memory.scan_processes	E-006	Pool scan (psscan) found 2212 EPROCESS structure Memory.raw. This includes terminated/hidden pro
F-005	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 17.57.144.165 (j APSDaemon.exe (PID 13224). Grouped connection
F-006	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.114.128.43 (j (PID 11672). Grouped connections to the same ext
F-007	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.114.142.209 Teams.exe (PID 11672). Grouped connections to th
F-008	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	5 established TCP connection(s) to 23.197.181.157 WinStore.App.e (PID 30012). Grouped connections
F-009	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.242.211.89 (j (PID 4872). Grouped connections to the same exte
F-010	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.113.194.132 SearchFilterHo (PID 24920). Grouped connections
F-011	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.114.132.120 Teams.exe (PID 11672). Grouped connections to th
F-012	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 13.107.136.9 (p OneDrive.exe (PID 9648). Grouped connections to
F-013	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.114.75.149 (j OneDrive.exe (PID 9648). Grouped connections to
F-014	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.179.224.121 OneDrive.exe (PID 6188). Grouped connections to
F-015	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 172.217.12.170 GoogleDriveFS. (PID 14832). Grouped connections
F-016	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	2 established TCP connection(s) to 213.202.233.10 svchost.exe (PID 1248). Grouped connections to th
F-017	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 81.30.144.115 (j svchost.exe (PID 1248). Grouped connections to th
F-018	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.242.211.89 (j OneDrive.exe (PID 9648). Grouped connections to
F-019	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	5 established TCP connection(s) to 23.46.190.35 (p WinStore.App.e (PID 30012). Grouped connections
F-020	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 17.248.138.42 (j ApplePhotoStre (PID 12888). Grouped connections
F-021	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 81.30.144.115 (j svchost.exe (PID 1248). Grouped connections to th

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-022	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 172.217.10.42 (google drivesyn (PID 8432). Grouped connections to
F-023	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 54.82.161.19 (p (PID 12808). Grouped connections to the same ext
F-024	ACTIVE	0.75	observation	T1071	memory.scan_network	E-007	1 established TCP connection(s) to 52.114.133.189 Teams.exe (PID 15636). Grouped connections to th
F-025	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	5 suspicious VAD region(s) in MsMpEng.exe (PID 4 possible shellcode injection. Protections: PAGE_EXI
F-026	ACTIVE	0.50	observation	T1055	memory.detect_injection	E-008	1 suspicious VAD region(s) in dllhost.exe (PID 8748 possible shellcode injection. Protections: PAGE_EXI
F-027	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	4 suspicious VAD region(s) in SearchApp.exe (PID 8 possible shellcode injection. Protections: PAGE_EXI
F-028	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	1 suspicious VAD region(s) in LockApp.exe (PID 978 possible shellcode injection. Protections: PAGE_EXI
F-029	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	1 suspicious VAD region(s) in RuntimeBroker. (PID — possible shellcode injection. Protections: PAGE_
F-030	ACTIVE	0.50	observation	T1055	memory.detect_injection	E-008	1 suspicious VAD region(s) in Teams.exe (PID 1563 possible shellcode injection. Protections: PAGE_EXI
F-031	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	2 suspicious VAD region(s) in SearchApp.exe (PID 1 — possible shellcode injection. Protections: PAGE_
F-032	ACTIVE	0.60	observation	T1055	memory.detect_injection	E-008	1 suspicious VAD region(s) in smartscreen.ex (PID — possible shellcode injection. Protections: PAGE_
F-033	ACTIVE	0.30	hypothesis	T1036	state.submit_finding	E-005	csrss.exe (PID 800) and winlogon.exe (PID 896) bot of expected smss.exe parent; most likely a PID-reu
F-034	ACTIVE	0.75	observation	T1014	state.submit_finding	E-006	No DKOM rootkit hiding confirmed: 11 PIDs found (LocalBridge.exe, RuntimeBroker, SDXHelper, Searc
F-035	ACTIVE	0.80	observation	T1055	state.submit_finding	E-008	No confirmed code injection: malfind returned 16 (MsMpEng.exe×5, dllhost.exe, SearchApp.exe×4, Lc
F-036	ACTIVE	0.50	observation	T1567	state.submit_finding	E-007	Full iCloud suite resident in memory at acquisition iCloudDrive.exe PID 13260, iCloudPhotos.exe PID ‘
F-037	ACTIVE	0.30	observation		memory.list_dlls	E-024	PID 13224: 83 DLLs loaded. All DLLs loaded from s
F-038	ACTIVE	0.30	observation		memory.list_dlls	E-025	PID 27740: 122 DLLs loaded. All DLLs loaded from
F-039	ACTIVE	0.30	observation		memory.list_dlls	E-026	PID 13260: 135 DLLs loaded. All DLLs loaded from
F-040	ACTIVE	0.30	observation		memory.list_dlls	E-027	PID 12756: 128 DLLs loaded. All DLLs loaded from
F-041	ACTIVE	0.85	observation	T1136.001	state.submit_finding	E-036	Unknown account 'srl-h' profile directory created 2 laptop. Account is not listed in any known SRL acco
F-042	ACTIVE	0.90	observation	T1070.001	state.submit_finding	E-037	Security event log archive created with zero bytes. break-in window). MFT EntryNumber 131916:...
F-043	ACTIVE	0.85	observation	T1021.001	state.submit_finding	E-037	RDP/Terminal Services session activity during brea newly-created RDP artifacts at 2020-11-14 03:42:50
F-044	ACTIVE	0.70	observation	T1567.002	state.submit_finding	E-037	OneDrive Business1 and Personal, and Google Dri synchronization throughout the break-in window (

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-045	ACTIVE	0.75	observation	T1567.002	state.submit_finding	E-035	Google Drive TempData directory created at 2020-break-in window, indicating a new Google Drive sy
F-046	ACTIVE	0.90	observation		disk.summarize_evtx	E-048	Parsed 133545 event log entries from /mnt/disk/W filter: Security). Events may reveal logon activity, p
F-047	ACTIVE	0.95	observation	T1110.003	state.submit_finding	E-056	NTLM brute force campaign: 58,713 EID 4776 (NTL (failed logon) events in Security.evtx spanning 202
F-048	ACTIVE	0.98	observation	T1021.001	state.submit_finding	E-062	INBOUND RDP takeover from attacker host 'cobra' 03:42:44 UTC (22:42 EST Nov 13) the LOCAL consol
F-049	ACTIVE	0.97	observation	T1021.001	state.submit_finding	E-061	Lateral movement from Fred's laptop to shieldbas UTC (00:00 EST) attacker used RDP Client to conne
F-050	ACTIVE	0.90	observation	T1114.001	state.submit_finding	E-063	Outlook email store (PST/OST) accessed via admin at 2020-11-14 14:01:22 UTC records SRL-FORGE\fre
F-051	ACTIVE	0.70	observation		disk.extract_usn_journal	E-072	USN Journal parsed: 383915 change records from . LAPTOP/artifacts/raw/usn/\$Usn\rnl_\$J. Use run_an
F-052	ACTIVE	0.95	observation	T1114.001	state.submit_finding	E-083	Attacker created backup.pst at 2020-11-14 13:39:2 Users\fredr\OneDrive\Documents\Outlook Files, w then deleted...
F-053	ACTIVE	0.80	observation	T1005	state.submit_finding	E-083	USN journal records StarFury.zip (BasicInfoChange Temp2_StarFury.zip and Temp3_StarFury.zip (FileCi
F-054	ACTIVE	0.90	observation	T1070.004	state.submit_finding	E-083	SDelete.zip (Sysinternals secure-deletion tool) was Zone.Identifier ADS written (NamedDataExtend Si
F-055	ACTIVE	0.95	observation		disk.extract_prefetch	E-088	Prefetch: parsed 190 .pf files from /mnt/disk/Wind SHELLEXPERIENCEHOST.EXE to SVCHOST.EXE. Use
F-056	ACTIVE	0.98	observation	T1070.004	state.submit_finding	E-099	SDELETE.EXE executed 7 times across two paths du UTC 2020-11-14). First 2 runs from...
F-057	ACTIVE	0.85	observation	T1005	state.submit_finding	E-088	Prefetch confirms attacker opened SRL intellectual 11-14 RDP session: ACRORD32.EXE (Adobe Acrobat
F-058	ACTIVE	0.68	observation		disk.get_amcache	E-103	Amcache: parsed 253 execution records from /mnt/disk/Windows/appcompat/Programs/Amcac (Temp/AppData/Downloads). Use...
F-059	ACTIVE	0.95	observation	T1136.001	state.submit_finding	E-106	Amcache confirms the srl-h account downloaded 5 7bcd946326b67f806b3db4595ede9fbdf29d0c36) to
F-060	ACTIVE	0.85	observation	T1021.001	state.submit_finding	E-107	Amcache records mstsc.exe (Microsoft Remote De 08:19:18 UTC and control.exe (Control Panel) at 12
F-061	ACTIVE	0.63	observation	T1070.006	disk.extract_shimcache	E-110	ShimCache (AppCompatCache) parsed from SYSTE AppCompatCacheParser. 200 entries loaded (max_ standard...
F-062	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 1: C:\\$WinREAgent\Sc 35DA76FCB204\dismhost.exe LastModified: 2019
F-063	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 6: C:\WINDOWS\SoftwareDistribution\Download\Inst LastModified: 2020-11-11 02:15:58 ...
F-064	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 7: C:\WINDOWS\SoftwareDistribution\Download\Inst LastModified: 2020-11-11 02:15:53 ...
F-065	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 8: C:\Users\fredr\AppDataDDA-BFE1-4AF40458976D\dismhost.exe LastMc

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-066	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 13: SIGN.MEDIA=E311 LastModified: 2020-11-08 03:35:02 Executed: Yes
F-067	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 15: c:\users\fredr\appdata\local\temp\GOOGLE_DRIVE LastModified: 2020-10-15 00:19:54 ...
F-068	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 16: C:\WINDOWS\Inst 2020-11-10 14:09:26 Executed: Yes. Path is outsc
F-069	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 18: C:\Users\fredr\AppData\Local\Temp\GUMF3B9.tm LastModified: 2020-11-10 14:09:06 Executed: Yes
F-070	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 19: C:\Users\fredr\AppData\Local\Temp\GUMF3B9.tm 2020-11-10 14:09:09 Executed: No. Path is...
F-071	ACTIVE	0.53	hypothesis	T1059	disk.extract_shimcache	E-110	ShimCache entry at position 20: C:\Users\fredr\Do LastModified: 2020-11-10 14:09:06 Executed: No.
F-072	ACTIVE	0.80	observation	T1567.002	state.submit_finding	E-115	ShimCache positions 11–12 confirm Google Drive l crashpad_handler.exe, version 43.0.8.0) was amon
F-073	ACTIVE	0.55	observation	T1091	state.submit_finding	E-115	ShimCache position 13 records SIGN.MEDIA=E311 (Executed=Yes, \$SI modification time 2020-11-08 0
F-074	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 2 row(s) targ DhcpIPAddress=0.0.0.0; DhcpSubnetMask=0.0.0.0.
F-075	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 12 row(s) tar EnableDHCP=1; EnableDHCP=1. Promotion: non_S
F-076	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 3 row(s) targ RegisterAdapterName=0; RegistrationEnabled=0. I
F-077	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 245 row(s) ta Samples: Multiple=Action: Allow Active: FALSE; Mul
F-078	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 363 row(s) ta Samples: Multiple=Action: Allow Active: TRUE; Mult
F-079	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 2 row(s) targ microsoft.windows.startmenuexperiencehost_cw5
F-080	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 2 row(s) targ microsoft.windows.search_cw5n1h2txyewy. Sampl
F-081	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 1 row(s) targ \device\harddiskvolume3\users\fredr\appdata\loc
F-082	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 1 row(s) targ \device\harddiskvolume3\users\fredr\appdata\roa
F-083	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 1 row(s) targ \device\harddiskvolume3\users\fredr\appdata\loc
F-084	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 1 row(s) targ \device\harddiskvolume3\windows\system32\cmd
F-085	ACTIVE	0.72	observation	T1543.003	disk.extract_registry_run_keys	E-119	Registry persistence group (services): 1 row(s) targ \device\harddiskvolume3\users\fredr\appdata\loc
F-086	ACTIVE	0.72	observation	T1547.004	disk.extract_registry_run_keys	E-119	Registry persistence group (winlogon_shell): 1 row LastUsedUsername=fredr. Promotion: rare_autosti
F-087	ACTIVE	0.72	observation	T1547.004	disk.extract_registry_run_keys	E-119	Registry persistence group (winlogon_shell): 1 row 567677750-276746561-1002. Samples:...

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-088	ACTIVE	0.72	observation	T1547.001	disk.extract_registry_run_keys	E-119	Registry persistence group (run): 1 row(s) targeting \users\fredr\appdata\local\microsoft\onedrive\on
F-089	ACTIVE	0.72	observation	T1547.001	disk.extract_registry_run_keys	E-119	Registry persistence group (run): 1 row(s) targeting \users\fredr\appdata\local\microsoft\teams\updat
F-090	ACTIVE	0.72	observation	T1547.001	disk.extract_registry_run_keys	E-119	Registry persistence group (run): 1 row(s) targeting h\appdata\local\microsoft\onedrive\onedrive.exe.
F-091	ACTIVE	0.92	observation	T1592	state.submit_finding	E-126	Registry NTUSER.DAT for srl-h records FirstLogonT srl-h's FIRST login to Fred's laptop, 24 days before
F-092	ACTIVE	0.85	observation	T1567.002	state.submit_finding	E-124	Registry Run keys in fredr's NTUSER.DAT (LastWrite BOTH Google Drive Sync (googledrivesync.exe /au
F-093	ACTIVE	0.90	observation	T1547.001	state.submit_finding	E-131	Registry suppressed.csv (793 rows of grouped/de-reviewed. No attacker-planted persistence, fileless
F-094	ACTIVE	0.90	observation	T1048	state.submit_finding	E-134	SRUM (System Resource Utilization Monitor) datab table was empty — no per-process bytes_sent/byt
F-095	ACTIVE	0.48	observation		disk.extract_shellbags	E-138	ShellBags: parsed 182 BagMRU entries across 2 pr RENDERED the folder; NOT that files inside were op
F-096	ACTIVE	0.95	observation	T1567.002	state.submit_finding	E-145	ShellBags confirm Google Drive File Stream was m browsed the following G: drive folders during the i
F-097	ACTIVE	0.88	observation	T1070.004	state.submit_finding	E-144	srl-h ShellBag (2020-11-01 22:18:49 UTC) shows Ex the attacker explicitly browsed the Windows Prefe
F-098	ACTIVE	0.85	observation	T1213.002	state.submit_finding	E-145	ShellBags from fredr's profile show the attacker br folders during the early attack hours (03:52-04:30
F-099	ACTIVE	0.70	observation		disk.extract_lnk_files	E-150	LNK files: parsed 142 shortcuts across 2 profile(s). referenced, NOT that a human clicked it; corrobor
F-100	ACTIVE	0.98	observation	T1567.002	state.submit_finding	E-153	LNK file SRL-EMAIL-EXPORT.lnk (SourceCreated 20: proves the Outlook PST export was placed on Goo
F-101	ACTIVE	0.97	observation	T1052.001	state.submit_finding	E-153	CRIMSON2 (F:\) USB drive was used as primary ph created during attack window 2020-11-14 03:46-0
F-102	ACTIVE	0.96	observation	T1005	state.submit_finding	E-153	LNK file analysis confirms the breadth of SRL intell during the 2020-11-14 attack. Under fredr's profile
F-103	ACTIVE	0.70	observation		disk.extract_jump_lists	E-158	Jump Lists: parsed 372 destination entries across 2 to the application (AppId) that referenced it; corrol
F-104	ACTIVE	0.97	observation	T1021.001	state.submit_finding	E-167	Jump List for Remote Desktop Connection (AppId: was used to RDP to "base-rd-08.shieldbase.lan" on
F-105	ACTIVE	0.99	observation	T1114.001	state.submit_finding	E-167	Jump Lists confirm Microsoft Outlook 2016 was us (backup.pst) and the Google Drive PST (SRL-EMAIL
F-106	ACTIVE	0.97	observation	T1213.002	state.submit_finding	E-167	Jump Lists confirm the attacker accessed Maria Hil documents through Fred's authenticated Office 36
F-107	ACTIVE	0.70	observation		disk.extract_browser_history	E-172	Browser history: parsed 1178 visit/download recor the browser PROCESS logged the event, NOT that
F-108	ACTIVE	0.99	observation	T1070.004	state.submit_finding	E-173	Browser history (Chrome) proves SDelete was sear during the 2020-11-14 attack session. At 13:37:37 I
F-109	ACTIVE	0.99	observation	T1213.002	state.submit_finding	E-173	Edge browser download record at 04:13:22 UTC pr SharePoint OneDrive was downloaded directly to t
F-110	ACTIVE	0.92	observation	T1564	state.submit_finding	E-173	Browser history reveals three distinct anti-forensic attack: (1) Decoy PDF downloads at 04:48:59-04:49

ID	STATUS	CONF	EVIDENCE	MITRE	TOOL	EXECUTION	DESCRIPTION
F-111	ACTIVE	0.70	observation		disk.extract_registry_fileaccess	E-178	Registry file-access: surfaced 590 entries (lastvisite recentdocs, runmru, typedpaths, userassist, wordv
F-112	ACTIVE	0.98	observation	T1083	state.submit_finding	E-181	WordWheelQuery registry key (NTUSER.DAT, LastW UTC) records 11 search terms typed into Windows
F-113	ACTIVE	0.70	observation		disk.extract_recycle_bin	E-184	Recycle Bin: parsed 19 \$I deletion record(s) across to the bin under a SID via the Explorer shell, NOT t
F-114	ACTIVE	0.99	observation	T1552.001	state.submit_finding	E-184	Recycle Bin entry \$IDNBREY.pst proves the BitLock OneDrive Desktop and deleted to the Recycle Bin i
F-115	ACTIVE	0.99	observation	T1070.004	state.submit_finding	E-184	Recycle Bin contains four post-exfiltration cleanup cleanup sequence: (1) backup.pst (\$IDNBREY.pst) c
F-116	ACTIVE	0.70	observation		disk.extract_powershell_history	E-188	PowerShell history: parsed 6 command line(s) acrc commands were ENTERED in an interactive consol
F-117	ACTIVE	0.92	observation	T1021.001	state.submit_finding	E-188	srl-h PowerShell history (mtime 2020-10-21T03:46: attempts to run sdelete64.exe via UNC tsclient pat
F-118	ACTIVE	0.92	observation	T1070.004	state.submit_finding	E-188	srl-h's PowerShell history records sequential execu and sdelete64.exe -nobanner -z -c C:, zeroing all fr
F-119	ACTIVE	0.59	observation		disk.extract_scheduled_tasks	E-191	Scheduled tasks: parsed 225 on-disk task definition REGISTERED with a given command/principal as o
F-120	ACTIVE	0.85	observation		state.submit_finding	E-191	Scheduled tasks review (225 tasks, E-191) found no tasks are standard Windows built-in, Microsoft, Ad
F-121	ACTIVE	0.90	observation		sigma_hunt	E-201	Chainsaw/Sigma hunt: 7 filtered rule hits across /mnt/disk/Windows/System32/winevt/Logs/Securi {'informational': 7}. ATT&CK...
F-122	ACTIVE	0.95	observation	T1070.001	state.submit_finding	E-201	Chainsaw Sigma scan (2,284 rules, E-201) against PowerShell.evtx, and Application.evtx produced 0 i
F-123	ACTIVE	0.97	inference	T1070.004	state.submit_finding	E-204	compare_disk_and_memory (E-204) confirms two € HIGH discrepancies for SDELETE.EXE (F-056/F-115)
F-124	CONFIRMED	0.95	inference	T1005	state.submit_finding	E-206	Temporal cluster 03:42:49-03:46:33 UTC 2020-11-1. EVTX+MFT+file_system+Prefetch, confidence 0.95)
F-125	CONFIRMED	0.97	inference	T1114.001	state.submit_finding	E-206	Temporal cluster 14:00:50-14:04:07 UTC 2020-11-1. EVTX+file_system+registry, confidence 0.90) confir
F-126	ACTIVE	0.80	observation		analyze_vss	E-210	VSS analysis of /evidence/srl-2018/rocba/disk/rocb This may indicate VSS was disabled (T1490: Inhibit
F-127	ACTIVE	0.95	observation	T1490	state.submit_finding	E-210	analyze_vss (E-210) found 0 Volume Shadow Copie absent, meaning pre-clearing Security.evtx recover

Recorded Hunting Hypotheses

Each hypothesis formed during the hunt and its verdict after testing against the evidence: proven (CONFIRMED), disproven (REFUTED), or inconclusive (SUSPENDED). "Linked finding IDs" are the F-NNN that proved, refuted, or materially informed the verdict.

HYPOTHESIS ID	ATTACK CLASS	VERDICT	LINKED FINDING IDS	MITRE
01KTJCM8EVCN629KMMCX2V7C81	pre_planned_physical_intrusion_staged_account (rank 1)	CONFIRMED - proven	F-046, F-047, F-057, F-059, F-117, F-118, F-101, F-102, F-124	-
01KTJCM8EVAZZC9GZ1Y16S5BNV	multi_channel_exfiltration_usb_cloud_lateral_movement (rank 2)	CONFIRMED - proven	F-048, F-049, F-050, F-100, F-101, F-102, F-104, F-105, F-106, F-108, F-109, F-110, F-112, F-115, F-124, F-125	-
01KTJCM8EV2X43RSP45X7XXDHQ	bitlocker_recovery_key_exfiltration (rank 3)	SUSPENDED - inconclusive	F-112, F-114	-
01KTJCM8EVYAD4MTFA8X3CZARX	icloud_opportunistic_cloud_sync_exfiltration (rank 4)	SUSPENDED - inconclusive	F-038	-
01KTJCM8EV2D5MX77J654SNF78	anti_forensics_evidence_destruction_multi_method (rank 5)	SUSPENDED - inconclusive	F-053, F-054, F-056, F-059, F-108, F-114, F-115, F-118, F-122, F-123	-

Analysis Lanes

LANE	STATUS	REQUIRED	AGENTS	EXECUTIONS	FINDINGS	SUMMARY
evidence_access	COMPLETE	yes	main-agent	0	0	load_memory: Memory dump ready at /evidence/memory/Rocba-Memory.raw
memory	COMPLETE	yes	main-agent	14	40	2186 processes enumerated; no DKOM hiding, no confirmed code injection (0 MZ headers); full iCloud suite active (APSDaemon, iCloudDrive, iCloudServices, iCloudIE) all loading...
disk_execution_persistence	COMPLETE	yes	main-agent	52	60	PowerShell history proves RDP tsclient access from rsydow's machine and sdelete64 free-space wiping; scheduled tasks show no attacker persistence; all mandatory file-access...
event_auth	COMPLETE_WITH_GAPS	yes	main-agent	22	6	EVTX analysis confirms full attacker lifecycle: 58K+ NTLM brute force (T1110.003), RDP session takeover from 'cobra' (52.249.198.56) at 22:42 EST (T1021.001), lateral movement...
anti_forensics_recovery	COMPLETE_WITH_GAPS	yes	main-agent	1	5	analyze_vss found 0 shadow copies (E-210) — VSS recovery of pre-clearing Security.evtx is impossible; F-127 documents the terminal evidence gap as a permanent irrecoverable...
timeline_correlation	COMPLETE_WITH_GAPS	yes	main-agent	25	14	Sigma scan (2,284 rules, E-201) yielded 0 medium/high/critical hits; Security.evtx had 12 corrupt chunks confirming log clearing — the negative Sigma result corroborates...
synthesis_corroboration	COMPLETE	yes	main-agent	2	3	compare_disk_and_memory (15 discrepancies, all explained) and find_temporal_clusters (5 clusters) promoted two CONFIRMED findings: active SRL IP collection at 03:42-03:46 UTC...

Orchestration Warnings

- None recorded.

Open Questions

- No open questions recorded.

Status Flags

```
{
  "open_leads": true,
  "anti_forensics_warning": false,
  "unresolved_discrepancy": true,
  "specialist_lanes_inferred": false,
  "hypotheses_unresolved": 0
}
```

Findings Status Breakdown

ACTIVE

125

CONFIRMED

2